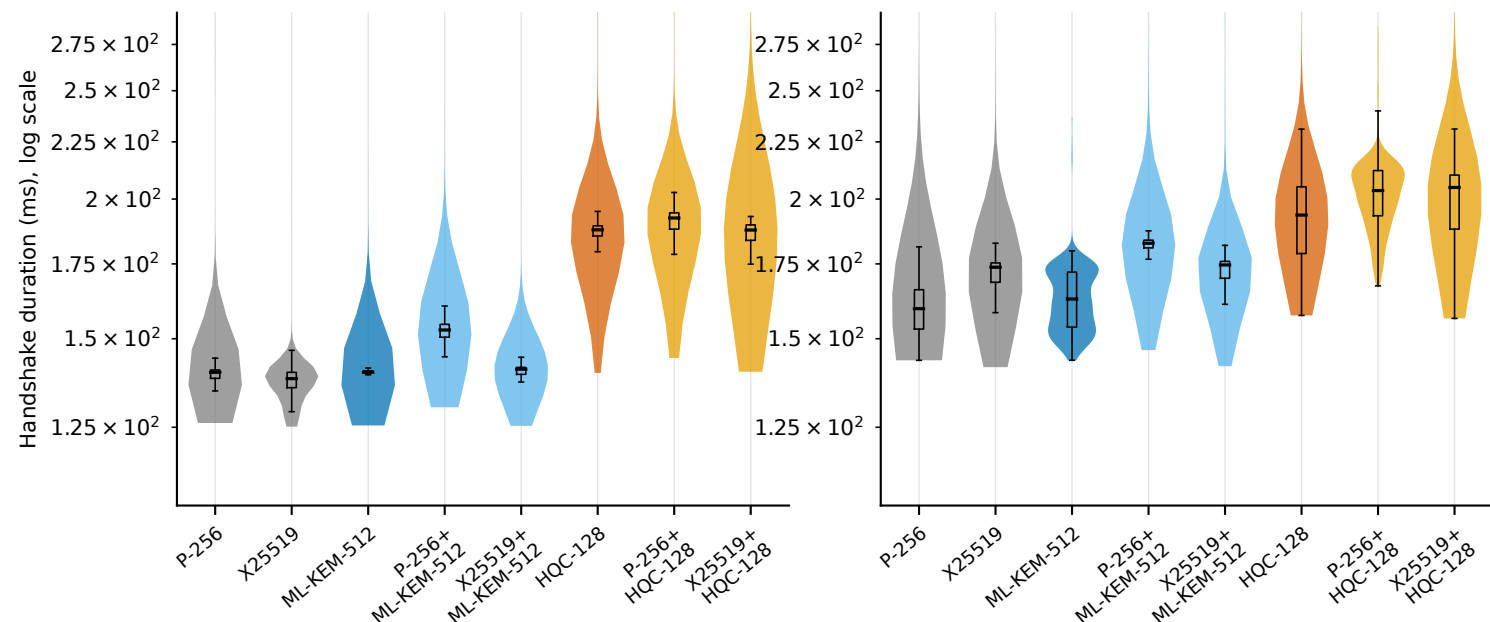


(a)

ed25519 (classical)

(b)

mldsa44 (post-quantum)

**(c)**

Classical KEX ML-KEM (pure) ML-KEM (hybrid) HQC (pure) HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	141.3	32.3	1.15k	169.9	55.4	1.19k
X25519	137.5	9.3	396.1	174.8	40.2	1.32k
ML-KEM-512	142.0	33.5	1.20k	163.2	6.8	236.7
P-256+ML-KEM-512	156.2	41.1	1.17k	186.7	37.3	1.31k
X25519+ML-KEM-512	144.1	25.6	533.7	174.2	30.2	1.31k
HQC-128	191.6	36.1	1.21k	196.2	34.2	1.23k
P-256+HQC-128	193.8	35.0	1.18k	202.6	7.7	285.8
X25519+HQC-128	197.7	59.6	1.34k	206.7	43.5	1.39k